```
−> 63 targets .........                        [4.6s, 1+, 15−]

.. Derive: CoLaus .............................................................
−> 63 targets .........                        [4.6s, 1+, 15−]


−> 63 targets .........                        [4.6s, 1+, 15−]

.. Derive Education Level ..
−> 63 targets .........                        [4.6s, 1+, 15−]


−> 63 targets .........                        [4.6s, 1+, 15−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6719 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20404.
! 27 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


−> 63 targets .........                        [4.6s, 1+, 15−]

.. Derive Alcohol Category ..
−> 63 targets .........                        [4.6s, 1+, 15−]


−> 63 targets .........                        [4.6s, 1+, 15−]
. Total rows: 20442
. conso_hebdo present (primary source): 18833 (92.1%)
. sumalco present (comparison source) : 11045 (54%)
. alcohol_category_conso derived : 18833 (92.1%)
. alcohol_category_sumalco derived: 11045 (54%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non−drinker
5161 27.4 2 Light 4107 21.8 3 Moderate 4192 22.3 4 Heavy 5373 28.5 5 NA 1609
8.5
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2014 18.2 2 Light 469 4.2 3 Moderate 866 7.8 4 Heavy 7696 69.7 5 NA
9397 85.1
. Source agreement (rows where both present, n = 10726):
. Agree : 4811 (44.9%)
. Conso higher : 336 (3.1%)
. Sumalco higher: 5579 (52%)
Breakdown of mismatches:

. conso: Non−drinker / sumalco: Light (n = 259)
−> 63 targets .........                        [4.6s, 1+, 15−]

. conso: Non−drinker / sumalco: Moderate (n = 284)
−> 63 targets .........                        [4.6s, 1+, 15−]

. conso: Non−drinker / sumalco: Heavy (n = 512)
−> 63 targets .........                        [4.6s, 1+, 15−]

. conso: Light / sumalco: Non−drinker (n = 160)
```

```
-> 63 targets .........                        [4.6s, 1+, 15-]

. conso: Light / sumalco: Moderate (n = 457)
-> 63 targets .........                        [4.6s, 1+, 15-]

. conso: Light / sumalco: Heavy (n = 1755)
-> 63 targets .........                        [4.6s, 1+, 15-]

. conso: Moderate / sumalco: Non-drinker (n = 71)
-> 63 targets .........                        [4.6s, 1+, 15-]

. conso: Moderate / sumalco: Light (n = 21)
-> 63 targets .........                        [4.6s, 1+, 15-]

. conso: Moderate / sumalco: Heavy (n = 2312)
-> 63 targets .........                        [4.6s, 1+, 15-]

. conso: Heavy / sumalco: Non-drinker (n = 61)
-> 63 targets .........                        [4.6s, 1+, 15-]

. conso: Heavy / sumalco: Light (n = 7)
-> 63 targets .........                        [4.6s, 1+, 15-]

. conso: Heavy / sumalco: Moderate (n = 16)
-> 63 targets .........                        [4.6s, 1+, 15-]


-> 63 targets .........                        [4.6s, 1+, 15-]

.. Derive Diabetes Status ..
-> 63 targets .........                        [4.6s, 1+, 15-]


-> 63 targets .........                        [4.6s, 1+, 15-]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 53 (0.3%)
. Sources available (1/2/3): 670 / 11725 / 7994
. Disagreement between any sources: 4.2%
. Disagreement between FPG and HbA1c: 1.9%
. Self-report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


-> 63 targets .........                        [4.6s, 1+, 15-]

.. Derive CVD Status ..
-> 63 targets .........                        [4.6s, 1+, 15-]


-> 63 targets .........                        [4.6s, 1+, 15-]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20422 | missing: 20
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17439 rows
```

```
-> 63 targets .........                        [4.6s, 1+, 15-]

.. Derive HRT Status ..
-> 63 targets .........                        [4.6s, 1+, 15-]


-> 63 targets .........                        [4.6s, 1+, 15-]
. hrt_status derived.
. Total rows: 20442 | derived: 9516 | missing (esthrp NA): 10926
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 5840
61.4 2 Past HRT 3 0 3 Current HRT 3673 38.6 4 NA 10926 115.


-> 63 targets .........                        [4.6s, 1+, 15-]

.. Physical Activity Derivation Report ..
-> 63 targets .........                        [4.6s, 1+, 15-]


-> 63 targets .........                        [4.6s, 1+, 15-]
. Tertile cut-points (from F1, N = 4089):
  Low < 168 | Medium 168-283 | High >= 283 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4089)
  Low 1357 (33.2%) Medium 1368 (33.5%) High 1364 (33.4%)
WHO corrected (N = 4089)
  Low 1419 (34.7%) Medium 1736 (42.5%) High 934 (22.8%)
. Visit: F2
Tertile (N = 2629)
  Low 869 (33.1%) Medium 928 (35.3%) High 832 (31.6%)
WHO corrected (N = 2629)
  Low 931 (35.4%) Medium 1153 (43.9%) High 545 (20.7%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 63 targets .........                        [4.6s, 1+, 15-]

.. Method alignment (tertile vs WHO corrected)
-> 63 targets .........                        [4.6s, 1+, 15-]
. Visit F1: 3397/4089 in same category (83.1%)
. Visit F2: 2172/2629 in same category (82.6%)


-> 63 targets .........                        [4.6s, 1+, 15-]

.. Dairy sub-category derivation ..
```

```
-> 63 targets .........                        [4.6s, 1+, 15-]


-> 63 targets .........                        [4.6s, 1+, 15-]
. derive_dairy: derived variables added.
  total: 10696
  fermented: 10795
  non-fermented: 10736
  low-fat: 10755
  high-fat: 10704
  total (excl. fondue): 10699
! .visit is an unordered factor. Using current factor level order.


-> 63 targets .........                        [4.6s, 1+, 15-]

.. Cumulative average dairy derivation ..
-> 63 targets .........                        [4.6s, 1+, 15-]


-> 63 targets .........                        [4.6s, 1+, 15-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 12502
-> 63 targets .........                        [4.6s, 1+, 15-]

  dairy_fermented_gday_cumavg: 12516
-> 63 targets .........                        [4.6s, 1+, 15-]

  dairy_non_fermented_gday_cumavg: 12504
-> 63 targets .........                        [4.6s, 1+, 15-]

  dairy_lowfat_gday_cumavg: 12511
-> 63 targets .........                        [4.6s, 1+, 15-]

  dairy_highfat_gday_cumavg: 12502
-> 63 targets .........                        [4.6s, 1+, 15-]

  dairy_total_nofondue_gday_cumavg: 12502
-> 63 targets .........                        [4.6s, 1+, 15-]


-> 63 targets .........                        [4.6s, 1+, 15-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 63 targets .........                        [4.6s, 1+, 15-]


-> 63 targets .........                        [4.6s, 1+, 15-]

  FFQ1amount_cumavg: 12538
-> 63 targets .........                        [4.6s, 1+, 15-]

  FFQ2amount_cumavg: 12524
-> 63 targets .........                        [4.6s, 1+, 15-]

  FFQ3amount_cumavg: 12533
```

```
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ4amount_cumavg: 12528
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ5amount_cumavg: 12527
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ6amount_cumavg: 12534
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ7amount_cumavg: 12543
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ8amount_cumavg: 12528
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ52amount_cumavg: 12524
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ53amount_cumavg: 12513
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ63amount_cumavg: 12523
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ68amount_cumavg: 12525
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ71amount_cumavg: 12516
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ82amount_cumavg: 12524
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ83amount_cumavg: 12522
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ84amount_cumavg: 12523
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ85amount_cumavg: 12521
-> 63 targets .........                          [4.6s, 1+, 15-]

  FFQ86amount_cumavg: 12524
-> 63 targets .........                          [4.6s, 1+, 15-]


-> 63 targets .........                          [4.6s, 1+, 15-]

.. Dairy protein content derivation ..
-> 63 targets .........                          [4.6s, 1+, 15-]


-> 63 targets .........                          [4.6s, 1+, 15-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 12502
```

```
−> 63 targets .........                         [4.6s, 1+, 15−]

.. Non−dairy protein cumulative average derivation ..
−> 63 targets .........                         [4.6s, 1+, 15−]


−> 63 targets .........                         [4.6s, 1+, 15−]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 12569
. prot_content_nondairy_cumavg valid: 12500


−> 63 targets .........                         [4.6s, 1+, 15−]

.. Derive Dairy Servings ..
−> 63 targets .........                         [4.6s, 1+, 15−]


−> 63 targets .........                         [4.6s, 1+, 15−]
. Total rows: 20442
. dairy_freq_total derived : 11027 (53.9%)
. dairy_portion_total derived : 11027 (53.9%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 8634 42.2 2 >= 2 servings/day 2393 11.7 3 NA 9415 46.1
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 9887 48.4 2 >= 2 servings/day 1140 5.6 3 NA 9415 46.1
. Method agreement (rows with both non−missing, n = 11027):
. Agree : 9774 (88.6%)
. Disagree: 1253 (11.4%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1253
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4605 472 10.2 2 F2 3719 465 12.5 3 F3 2703 316 11.7


−> 63 targets .........                         [4.6s, 1+, 15−]

.. Derive Dairy Quartiles ..
−> 63 targets .........                         [4.6s, 1+, 15−]


−> 63 targets .........                         [4.6s, 1+, 15−]
. Baseline quartile cut−points (g/day, n = 4605):
. Q1 | Q2 boundary : 110.4
. Q2 | Q3 boundary : 196.8
. Q3 | Q4 boundary : 302.2
. Overall quartile cut−points (g/day, n = 12502):
. Q1 | Q2 boundary : 116.5
. Q2 | Q3 boundary : 195
. Q3 | Q4 boundary : 295.8
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
```

```
Baseline NA 6733 100  6 F1 Q1 1152 22.7  7 F1 Q2 1151 22.7  8 F1 Q3 1151 22.7  9 F1
Q4 1151 22.7 10 F1 NA 459 9.1 11 F2 Q1 979 20 12 F2 Q2 1202 24.6 13 F2 Q3 1137
23.2 14 F2 Q4 1037 21.2 15 F2 NA 539 11 16 F3 Q1 755 20.1 17 F3 Q2 1073 28.6 18
F3 Q3 906 24.2 19 F3 Q4 808 21.5 20 F3 NA 209 5.6
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1231 24.3  7 F1 Q2 1048 20.7  8 F1 Q3 1121 22.1  9 F1
Q4 1205 23.8 10 F1 NA 459 9.1 11 F2 Q1 1067 21.8 12 F2 Q2 1089 22.3 13 F2 Q3
1119 22.9 14 F2 Q4 1080 22.1 15 F2 NA 539 11 16 F3 Q1 829 22.1 17 F3 Q2 987
26.3 18 F3 Q3 885 23.6 19 F3 Q4 841 22.4 20 F3 NA 209 5.6
. Dairy quartiles derived.
. dairy_quartile_baseline: 12502 / 20442 rows non−missing
. dairy_quartile_overall : 12502 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  12502):
. Same quartile : 12070 (96.5%)
. Different quartile: 432 (3.5%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 241 2 Q2 Q3 61 3 Q3 Q4 130
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4605 157 3.4 2 F2 4355 156 3.6 3 F3 3542 119 3.4


−> 63 targets .........                        [4.6s, 1+, 15−]

.. Derive ATCs ..
−> 63 targets .........                        [4.6s, 1+, 15−]


−> 63 targets .........                        [4.6s, 1+, 15−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 63 targets .........                        [4.6s, 1+, 15−]

.. Derive Smoking Status ..
−> 63 targets .........                        [4.6s, 1+, 15−]


−> 63 targets .........                        [4.6s, 1+, 15−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 18926 2
derived 698 3 NA 818


−> 63 targets .........                        [4.6s, 1+, 15−]

.. Derive BMI ..
−> 63 targets .........                        [4.6s, 1+, 15−]


−> 63 targets .........                        [4.6s, 1+, 15−]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 19759 | missing: 683
. BMI range (non−missing): 13.7 − 79.7 kg/m²
```

```
. Mean ± SD: 26.1 ± 4.6 kg/m²
. Agreement with BMI_source (rows with both non−missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01−0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


−> 63 targets .........                        [4.6s, 1+, 15−]

.. Deriving BMI category ..
−> 63 targets .........                        [4.6s, 1+, 15−]


−> 63 targets .........                        [4.6s, 1+, 15−]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (683 missing BMI).
  Prevalence of Obesity: 17.4%
  Prevalence of Overweight + Obesity: 55.7%


−> 63 targets .........                        [4.6s, 1+, 15−]

.. Derive Age ..
−> 63 targets .........                        [4.6s, 1+, 15−]


−> 63 targets .........                        [4.6s, 1+, 15−]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non−missing): 34.9 − 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non−missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1−1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


−> 63 targets .........                        [4.6s, 1+, 15−]

.. Food group derivation (updated MSM structure) ..
−> 63 targets .........                        [4.6s, 1+, 15−]


−> 63 targets .........                        [4.6s, 1+, 15−]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810
```

```
-> 63 targets .........                        [4.6s, 1+, 15-]

.. Derive htn Status ..
-> 63 targets .........                        [4.6s, 1+, 15-]


-> 63 targets .........                        [4.6s, 1+, 15-]
. htn_status derived.
. Total rows: 20442 | Yes: 8720 | No: 11709 | Missing: 13
. Prevalence: 42.7% (among non-missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8720
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self-reported, tier 3) : 0
```